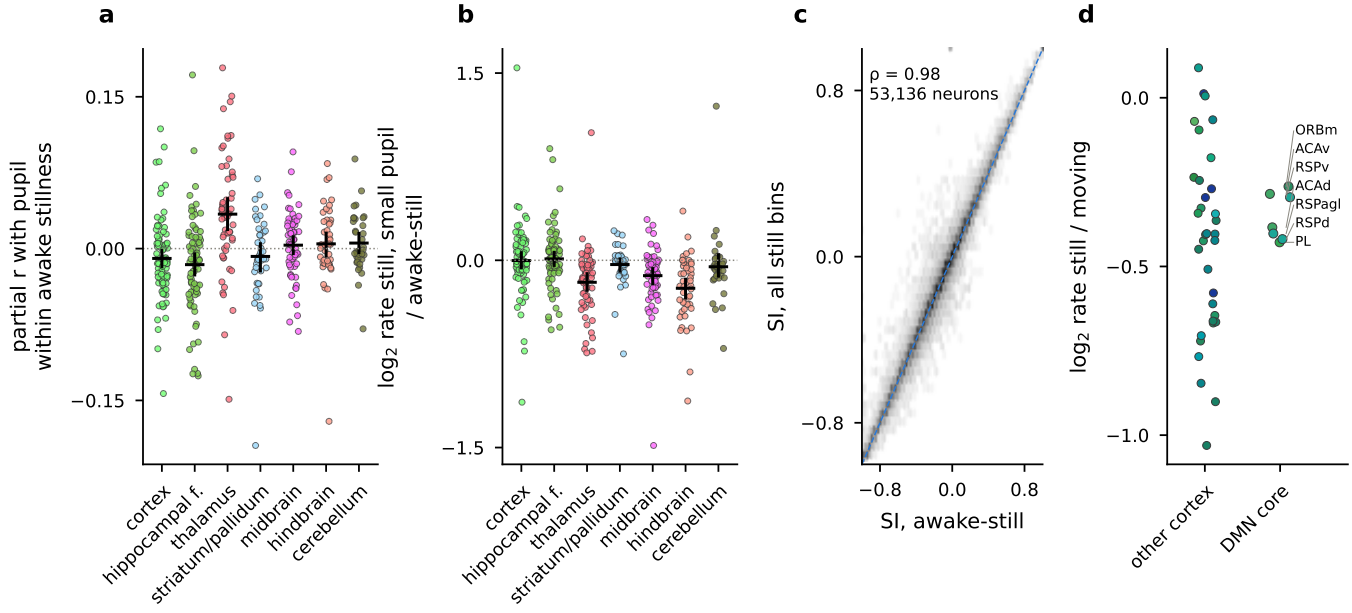


Figure 6: Arousal within stillness, the sleep gate and the DMN core. **a** Partial correlation of each neuron's 1 s spike count with pupil diameter within awake stillness (time and motion energy regressed out); dots are mice. **b** Rate in still bins with a pupil below the gate (but $\leq$30 s since movement) relative to awake stillness; lower in thalamus, midbrain and hindbrain, unchanged in cortex and hippocampus. **c** Stillness index with and without the anti-sleep gate. **d** Cortical regions: conservative DMN core vs other cortex (core -0.35, other -0.43, region permutation p = 0.49); dots in Allen atlas region colours.